

Sharp Real-Axis Decay of the Fourier Transform of Rvachev's Up-Function

Dyadic dynamics, Gelfond extremals, Lambert W boundary layers, and two-adic
troughs

A rigorous analysis of the decay-rate conjecture
in Vladimir Reshetnikov's `ProveIt` research frontier

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Abstract

With the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i t \xi} dt$, the normalized Rvachev up-function has Fourier transform

$$\Phi(\xi) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi \xi}{2^h}\right), \quad \operatorname{sinc} z = \frac{\sin z}{z}.$$

The product is rapidly decreasing, but it has a highly nonuniform real-axis amplitude: it vanishes at every nonzero integer, its dyadic orbits are governed by Birkhoff sums of $\log |\sin \pi x|$, and its largest peaks in a dyadic annulus arise from a boundary layer that must first escape the zero at the left endpoint and then shadow the maximizing Thue–Morse orbit $\{1/3, 2/3\}$.

This article gives a rigorous hierarchy of decay laws. Universally,

$$\limsup_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\frac{1}{2 \log 2}, \quad \liminf_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\infty.$$

For almost every fixed dyadic ray $x = 2^k y$, $1 < y < 2$, an exact second term is obtained from Birkhoff's theorem. More sharply, put

$$M_k = \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|, \quad \lambda = \log 2, \quad \beta = \log \frac{\sqrt{3}}{2}, \quad c = \frac{\lambda}{\pi} \sqrt{\frac{3}{2}},$$

and let W be the principal Lambert function. We prove

$$\begin{aligned} \log M_k = & -\frac{\lambda}{2} k(k+1) - (k+1) \log \pi + (k+1) \beta \\ & - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2\lambda} + \mathcal{O}((\log \log k)^2). \end{aligned}$$

Thus the envelope contains a second log-normal suppression in $\log \log x$; it is not a log-normal term multiplied merely by a fixed power of $\log x$. An elementary algebraic subaction proves the needed sharp Gelfond exponent. Finally, if E_n is the unique interzero peak on $(n, n+1)$, then for each fixed odd positive integer m ,

$$E_{m2^k} \sim \frac{H(1)|H(m)|}{e(k+1)} (m2^k)^{-(k+1)}, \quad H(z) = \prod_{r=1}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^r}\right).$$

These two-adically exceptional peaks are exponentially smaller on the log-normal scale than the largest peak in the same dyadic annulus. The results show why no positive smooth normalization can make the pointwise oscillation have a nonzero constant amplitude, and they separate pointwise, typical, extremal, and averaged notions of decay.

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1 The question and the necessary distinction between decay notions

Rvachev's up-function is an even, compactly supported C^∞ bump. Under the Fourier convention used above, its Fourier transform is the entire function

$$\Phi(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right). \quad (1)$$

This product already occurs in the classical construction by Jessen and Wintner and in the later theory of the Fabius and Rvachev functions; a modern self-contained treatment is given by Arias de Reyna [1]. The product representation, its cosine form, and its canonical zero product are also summarized in the Fabius-function literature [8].

The draft that motivated the present analysis [7] proposes a single asymptotic of the schematic form

$$|\Phi(x)| \asymp (\log x)^{-p} x^{-\log(2\pi)/\log 2} \exp\left(-\frac{(\log x)^2}{2\log 2}\right),$$

and suggests that a reciprocal normalization should leave an oscillation of asymptotically constant amplitude. The leading quadratic logarithm is in the right family, but the statement cannot be correct as a pointwise equivalent: Φ vanishes at every nonzero integer, and its nonzero peaks vary by an additional log-normal factor even inside a single dyadic annulus. Moreover, the extremal envelope has a Lambert W correction of size $-\frac{1}{2\log 2}(\log \log x)^2$, far larger than a fixed power of $\log x$.

The word “amplitude” can therefore refer to several inequivalent objects. We shall distinguish the following.

1. The universal pointwise logarithmic scale of $\Phi(x)$.
2. The decay along a fixed dyadic ray $x = 2^k y$.
3. The largest value

$$M_k = \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|$$

in a dyadic annulus.

4. The unique local peak

$$E_n = \max_{n \leq x \leq n+1} |\Phi(x)|$$

between two consecutive integer zeros.

5. Integral or Cesaro averages, which depend on L^1 , rather than L^∞ , information about Thue–Morse products.

The first four are treated rigorously here. The fifth is reduced to a separate transfer-operator problem in [section 9](#).

Main verdict. The correct leading scale is log-normal: $-\log |\Phi(x)|$ is of order $(\log x)^2$ at its large nonzero peaks. There is, however, no single pointwise asymptotic. Typical dyadic rays, maximizing rays, dyadic-annulus maxima, and peaks following highly even integers have different lower-order laws. The sharp dyadic-annulus envelope is the Lambert W formula in [theorem 6.4](#); its expansion begins with a second quadratic logarithm in $\log \log x$.

Throughout the article,

$$\lambda = \log 2, \quad \beta = \log \frac{\sqrt{3}}{2} = \frac{1}{2} \log 3 - \log 2 < 0, \quad \log 0 = -\infty. \quad (2)$$

All logarithms are natural.

2 Product identities, zeros, and one peak per unit interval

2.1 Entire convergence and the scaling law

Since $\operatorname{sinc} w = 1 - w^2/6 + \mathcal{O}(w^4)$ near the origin, the product in (1) converges locally uniformly on $\mathbb{C}$. It therefore defines an even entire function. Shifting the index gives the exact refinement law

$$\Phi(2z) = \operatorname{sinc}(2\pi z)\Phi(z), \quad \Phi(z) = \operatorname{sinc}(\pi z)\Phi(z/2). \quad (3)$$

Repeated use of the first identity yields

$$\Phi(2^k y) = \Phi(y) \prod_{j=1}^k \operatorname{sinc}(\pi 2^j y). \quad (4)$$

This elementary formula is the bridge from the Fourier transform to the doubling map.

Define the tail product

$$H(z) = \prod_{r=1}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^r}\right). \quad (5)$$

Then

$$\Phi(z) = \operatorname{sinc}(\pi z)H(z). \quad (6)$$

The function H is entire. It is positive on $[0, 2)$, and

$$h := H(1) = -\Phi'(1) = 0.55377127588881009534\dots \quad (7)$$

The numerical value in (7) is included only for reference; none of the proofs is numerical.

2.2 Canonical zero product

Euler's product

$$\frac{\sin \pi z}{\pi z} = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)$$

can be inserted into every factor of (1). Reindexing the locally uniformly convergent product gives

$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+v_2(m)}, \quad (8)$$

where $v_2(m)$ is the exponent of 2 in m . Indeed, a fixed integer m occurs once for every h with $2^h \mid m$, hence with multiplicity $1 + v_2(m)$. Formula (8) appears in Arias de Reyna's treatment [1].

Consequently, the real zeros are precisely the nonzero integers, and the zero at $m \neq 0$ has multiplicity

$$\operatorname{ord}_m \Phi = 1 + v_2(|m|). \quad (9)$$

The multiplicities are the first indication that powers of two are exceptional.

2.3 Uniqueness of the interzero extremum

The canonical product gives more than the zero set. It implies that every open unit interval contains exactly one critical point of Φ , hence exactly one peak of its absolute value.

Theorem 2.1 (One peak between consecutive integer zeros). *For every integer $n \geq 1$, there is a unique point $c_n \in (n, n+1)$ at which $\Phi'(c_n) = 0$. The function $|\Phi|$ is strictly increasing on (n, c_n) and strictly decreasing on $(c_n, n+1)$. In particular,*

$$E_n := |\Phi(c_n)| = \max_{n \leq x \leq n+1} |\Phi(x)|$$

is well defined.

Proof. Away from the integer zeros, logarithmic differentiation of (8) is justified by locally uniform convergence and gives

$$\frac{\Phi'(x)}{\Phi(x)} = -2x \sum_{m=1}^{\infty} \frac{1 + v_2(m)}{m^2 - x^2}. \quad (10)$$

Put

$$S(x) = \sum_{m=1}^{\infty} \frac{1 + v_2(m)}{m^2 - x^2}.$$

On $(n, n+1)$, termwise differentiation gives

$$S'(x) = 2x \sum_{m=1}^{\infty} \frac{1 + v_2(m)}{(m^2 - x^2)^2} > 0.$$

The term $m = n$ forces $S(x) \rightarrow -\infty$ as $x \downarrow n$, while the term $m = n+1$ forces $S(x) \rightarrow +\infty$ as $x \uparrow n+1$. Thus S has one and only one zero. By (10), this is the unique zero of Φ'/Φ . Its sign changes from positive to negative, so $\log |\Phi|$, and hence $|\Phi|$, first increases and then decreases. $\square$

The theorem provides a stable numerical procedure: maximize the logarithm of the product separately on each unit interval, then take the largest peak in the desired dyadic annulus. It also makes the contrast in [section 7](#) unambiguous: the exceptional small quantities there are genuine local maxima, not samples accidentally taken near a zero.

3 The universal leading logarithmic scale

A first estimate already determines the coefficient of $(\log x)^2$. It is sharp in the limsup sense but cannot be a pointwise equivalent because of the zeros.

Theorem 3.1 (Sharp universal leading coefficient). *As $x \rightarrow +\infty$,*

$$\limsup_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\frac{1}{2\lambda}, \quad \liminf_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\infty. \quad (11)$$

Proof. Let $2^k \leq x < 2^{k+1}$. For $0 \leq h \leq k$, use $|\sin u| \leq 1$, while for $h > k$ use $|\operatorname{sinc} u| \leq 1$. Then

$$|\Phi(x)| \leq \prod_{h=0}^k \frac{2^h}{\pi x} = \frac{2^{k(k+1)/2}}{(\pi x)^{k+1}}. \quad (12)$$

Because $k = (\log x)/\lambda + \mathcal{O}(1)$, taking logarithms gives

$$\log |\Phi(x)| \leq -\frac{(\log x)^2}{2\lambda} + \mathcal{O}(\log x),$$

which proves the upper bound for the limsup.

For the reverse inequality, take $x_k = 2^k/3$. Iterating (3) and using $|\sin(\pi 2^j/3)| = \sqrt{3}/2$ gives the exact identity

$$|\Phi(2^k/3)| = |\Phi(1/3)| \left(\frac{3\sqrt{3}}{2\pi} \right)^k 2^{-k(k+1)/2}. \quad (13)$$

Since $\log(2^k/3) = k\lambda + \mathcal{O}(1)$, this sequence attains the coefficient $-1/(2\lambda)$. Finally, $\Phi(n) = 0$ for every positive integer n , so the $\liminf$ is $-\infty$. $\square$

Corollary 3.2 (Between polynomial and stretched-exponential decay). *The function Φ is rapidly decreasing: for every $N > 0$, $|\Phi(x)| = \mathcal{O}_N(x^{-N})$. On the other hand, for every $\delta > 0$, it is not true that $|\Phi(x)| = \mathcal{O}(e^{-x^\delta})$ along all sufficiently large real x ; the sequence (13) is much larger than any such stretched exponential.*

Proof. The first assertion follows immediately from (12) by choosing more factors than a prescribed power, and is also proved directly in [1]. The second follows by comparing $(\log x)^2$ with x^δ along (13). $\square$

The theorem answers only the leading logarithmic question. The coefficient of $\log x$, and even the meaning of a second term, depends on how infinity is approached. The doubling dynamics makes that dependence explicit.

4 Dyadic rays and Thue–Morse products

4.1 Exact dynamical decomposition

Fix $y = 1 + \theta \in (1, 2)$. Since 2^j is even for $j \geq 1$, $\sin(\pi 2^j y) = \sin(\pi 2^j \theta)$. Taking logarithms in (4) gives the exact formula

$$\begin{aligned} \log |\Phi(2^k y)| &= \log |\Phi(y)| - \frac{\lambda}{2} k(k+1) - k \log(\pi y) \\ &\quad + \sum_{j=1}^k \log |\sin(\pi 2^j \theta)|. \end{aligned} \quad (14)$$

Thus every second-order statement is a statement about Birkhoff sums of the integrable, unbounded potential

$$\varphi(t) = \log |\sin \pi t| \quad (15)$$

under the doubling map $T(t) = 2t \pmod{1}$.

For later use, define the finite sine product

$$Q_n(t) = \prod_{j=0}^{n-1} |\sin(\pi 2^j t)|. \quad (16)$$

It is exactly the modulus of a Thue–Morse polynomial, up to the factor 2^{-n} . Indeed, if $s_2(m)$ is the sum of the binary digits of m , then

$$\prod_{j=0}^{n-1} (1 - e^{2\pi i 2^j t}) = \sum_{m=0}^{2^n-1} (-1)^{s_2(m)} e^{2\pi i m t}, \quad (17)$$

and therefore

$$2^n Q_n(t) = \left| \sum_{m=0}^{2^n-1} (-1)^{s_2(m)} e^{2\pi i m t} \right|. \quad (18)$$

The corresponding uniform exponent is the classical Gelfond exponent. Its special value needed here is proved elementarily in [section 5](#); the general ergodic-optimization framework is developed by Fan, Schmeling, and Shen [\[3\]](#).

4.2 A general ray law

Theorem 4.1 (Decay along a dyadic ray). *Let $y = 1 + \theta \in (1, 2)$, and suppose that the finite limit*

$$\alpha(\theta) = \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{j=1}^k \log |\sin(\pi 2^j \theta)| \quad (19)$$

exists. Then, with $x_k = 2^k y$,

$$\log |\Phi(x_k)| = -\frac{(\log x_k)^2}{2\lambda} + \left(\frac{\alpha(\theta)}{\lambda} - \frac{1}{2} - \frac{\log \pi}{\lambda} \right) \log x_k + o(\log x_k). \quad (20)$$

Proof. Insert $\sum_{j=1}^k \log |\sin(\pi 2^j \theta)| = k\alpha(\theta) + o(k)$ into [\(14\)](#). This gives

$$\log |\Phi(2^k y)| = -\frac{\lambda}{2} k^2 + k \left(\alpha(\theta) - \frac{\lambda}{2} - \log \pi - \log y \right) + o(k).$$

Substitute $k = (\log x_k - \log y)/\lambda$. The two terms containing $(\log y) \log x_k$ cancel, leaving [\(20\)](#). $\square$

The cancellation explains why the coefficient in terms of the actual frequency x_k , rather than the lower endpoint 2^k , is independent of the chosen mantissa y . The orbit average $\alpha(\theta)$ is the only remaining dynamical datum.

4.3 The almost-everywhere ray law

The potential φ in [\(15\)](#) belongs to $L^1([0, 1])$, and

$$\int_0^1 \log |\sin \pi t| dt = -\log 2 = -\lambda. \quad (21)$$

The doubling map is ergodic for Lebesgue measure, so Birkhoff's theorem applies.

Corollary 4.2 (Typical dyadic ray). *For Lebesgue-almost every $y \in (1, 2)$, as $k \rightarrow \infty$,*

$$\log |\Phi(2^k y)| = -\frac{\lambda}{2} k^2 - k \left(\frac{3\lambda}{2} + \log(\pi y) \right) + o(k). \quad (22)$$

Equivalently, for $x_k = 2^k y$,

$$\log |\Phi(x_k)| = -\frac{(\log x_k)^2}{2\lambda} - \left(\frac{3}{2} + \frac{\log \pi}{\lambda} \right) \log x_k + o(\log x_k). \quad (23)$$

The numerical value of the linear exponent is

$$\frac{3}{2} + \frac{\log \pi}{\log 2} = 3.151496129472318 \dots$$

This almost-everywhere law is a natural interpretation of a “typical” amplitude. It is not an envelope: exceptional orbits can have a much larger Birkhoff average and therefore a larger Fourier amplitude.

5 An elementary sharp Gelfond bound

The maximal possible orbit average in [theorem 4.1](#) is $\beta = \log(\sqrt{3}/2)$. This is the $q = 2, c = 1/2$ Gelfond exponent in the terminology of [3]. For the present problem it admits a short, fully explicit proof.

Theorem 5.1 (Uniform sine-product bound). *For every $t \in \mathbb{R}$ and every integer $n \geq 1$,*

$$Q_n(t) \leq \sqrt{\frac{5}{3}} \left(\frac{\sqrt{3}}{2} \right)^n. \quad (24)$$

The exponential rate is sharp: at $t = 1/3$ or $2/3$,

$$Q_n(t) = \left(\frac{\sqrt{3}}{2} \right)^n. \quad (25)$$

Consequently,

$$\sup_{t \in \mathbb{R}} \limsup_{n \rightarrow \infty} \frac{1}{n} \log Q_n(t) = \beta. \quad (26)$$

Proof. Set

$$u_j = \sin^2(\pi 2^j t).$$

The double-angle formula gives the logistic recursion

$$u_{j+1} = 4u_j(1 - u_j). \quad (27)$$

Introduce the positive affine function

$$V(u) = 1 + \frac{2}{3}u.$$

A direct factorization yields

$$3V(u) - 4uV(4u(1 - u)) = \frac{(2u + 1)(4u - 3)^2}{3} \geq 0 \quad (0 \leq u \leq 1). \quad (28)$$

Hence

$$2\sqrt{u_j} \leq \sqrt{3} \left(\frac{V(u_j)}{V(u_{j+1})} \right)^{1/2}.$$

Multiplying from $j = 0$ to $n - 1$ telescopes:

$$\prod_{j=0}^{n-1} 2|\sin(\pi 2^j t)| \leq 3^{n/2} \left(\frac{V(u_0)}{V(u_n)} \right)^{1/2} \leq 3^{n/2} \sqrt{\frac{5}{3}}.$$

Dividing by 2^n proves (24).

At $t = 1/3$, the doubling orbit alternates between $1/3$ and $2/3$, so every $u_j = 3/4$. Equality holds in the exponential rate and (25) follows. $\square$

Remark 5.2 (A calibrated subaction). Taking logarithms in the proof gives the pointwise inequality

$$\log |\sin \pi t| \leq \beta + \frac{1}{2} \log V(\sin^2 \pi t) - \frac{1}{2} \log V(\sin^2 2\pi t). \quad (29)$$

It is an explicit calibrated subaction for the maximizing two-cycle. The factorization (28) also identifies the equality set: apart from zero factors, equality occurs exactly when $\sin^2 \pi t = 3/4$, that is, on the orbit $\{1/3, 2/3\}$.

Corollary 5.3 (Maximizing dyadic rays). *The largest finite value of $\alpha(\theta)$ in (19) is β . It is attained by $\theta = 1/3, 2/3$ and by their eventually periodic preimages. Along such a maximizing ray, [theorem 4.1](#) has linear coefficient*

$$\frac{\beta}{\lambda} - \frac{1}{2} - \frac{\log \pi}{\lambda} = -\frac{a}{\lambda}, \quad a := \frac{\lambda}{2} + \log \pi - \beta = \log \left(\frac{2^{3/2} \pi}{\sqrt{3}} \right). \quad (30)$$

Numerically,

$$a = 1.635144512355263 \dots, \quad \frac{a}{\lambda} = 2.359014879111741 \dots$$

For example, taking $y = 4/3$, so that $\theta = 1/3$, gives the exact identity

$$|\Phi(2^k \frac{4}{3})| = |\Phi(\frac{4}{3})| \left(\frac{\sqrt{3}}{2\pi(4/3)} \right)^k 2^{-k(k+1)/2}. \quad (31)$$

The dyadic-annulus maximum is subtler. A fixed maximizing mantissa such as $4/3$ is separated by a fixed proportion from the lower endpoint 2^k , and the denominator $(1 + \theta)^{k+1}$ then incurs an exponential loss in k . The true maximum moves toward the endpoint while spending a slowly growing number of doublings reaching the maximizing cycle. That competition is the source of Lambert W .

6 The sharp dyadic-annulus envelope

This section contains the main asymptotic result. The proof has three components:

1. an exact reduction of Φ on $[2^k, 2^{k+1}]$ to a weighted Thue–Morse sine product;
2. an upper bound from [theorem 5.1](#) and a matching lower construction using dense preimages of the maximizing two-cycle;
3. a two-variable optimization whose logarithmic coordinates tile the real line and reduce to the principal Lambert W function.

6.1 Exact annular reduction

Write

$$x = 2^k(1 + \theta), \quad 0 \leq \theta \leq 1, \quad n = k + 1.$$

Using (6) for the factor at scale 2^k and then the first k refinement factors gives

$$|\Phi(2^k(1 + \theta))| = \frac{2^{-k(k+1)/2}}{\pi^n} H(1 + \theta)(1 + \theta)^{-n} Q_n(\theta). \quad (32)$$

Indeed, the numerator contains the factors $|\sin(\pi 2^j \theta)|$ for $0 \leq j \leq k$, and the denominators multiply to $\pi^n (1 + \theta)^n 2^{k(k+1)/2}$. Thus, if

$$A_n = \sup_{0 \leq \theta \leq 1} H(1 + \theta)(1 + \theta)^{-n} Q_n(\theta), \quad (33)$$

then

$$M_k = \pi^{-n} 2^{-k(k+1)/2} A_n. \quad (34)$$

All of the nontrivial lower-order behavior is contained in A_n .

6.2 Transition coordinates near the endpoint

The factor $(1 + \theta)^{-n}$ forces a maximizer toward $\theta = 0$, while $Q_n(0) = 0$. For $0 < \theta \leq 1/2$, choose an integer $r \geq 0$ and $s \in I := [1/4, 1/2]$ such that

$$\theta = 2^{-r}s. \quad (35)$$

Endpoint ambiguities are harmless. The integer r is the number of initial doublings spent escaping the zero at 0. If $r < n$, then

$$Q_n(2^{-r}s) = P_r(s)Q_{n-r}(s), \quad P_r(s) = \prod_{h=1}^r \sin\left(\frac{\pi s}{2^h}\right). \quad (36)$$

The small block has the uniform factorization

$$P_r(s) = (\pi s)^r 2^{-r(r+1)/2} H_r(s), \quad H_r(s) = \prod_{h=1}^r \operatorname{sinc}\left(\frac{\pi s}{2^h}\right). \quad (37)$$

On the compact interval I , the quantities $H_r(s)$ are bounded above and below by positive absolute constants, uniformly in r . Hence

$$\log P_r(s) = -\frac{\lambda}{2}r^2 + r \left(\log(\pi s) - \frac{\lambda}{2} \right) + \mathcal{O}(1) \quad (38)$$

uniformly for $s \in I$.

The tail estimate from [theorem 5.1](#) gives

$$Q_m(s) \leq C_G e^{m\beta}, \quad C_G = \sqrt{\frac{5}{3}}. \quad (39)$$

To match it from below while retaining freedom to choose s , we use preimages of the maximizing cycle.

Lemma 6.1 (Dense near-maximizing preimages). *There is an absolute constant $C > 0$ with the following property. For every $s_0 \in I$ and every integer $q \geq 1$, one can choose $s_q \in I$ such that*

$$|s_q - s_0| \leq 2^{-q}, \quad T^q(s_q) \in \left\{ \frac{1}{3}, \frac{2}{3} \right\}, \quad (40)$$

and, for every $m \geq q$,

$$Q_m(s_q) \geq \exp(m\beta - Cq^2). \quad (41)$$

Proof. The q -fold preimages of $1/3$ are

$$\frac{j + 1/3}{2^q}, \quad 0 \leq j < 2^q,$$

and have spacing 2^{-q} . Using these and, near an endpoint if needed, the preimages of $2/3$, choose $s_q \in I$ satisfying (40).

For $0 \leq j < q$, the point $T^j(s_q)$ is a nonintegral rational whose reduced denominator divides $3 \cdot 2^{q-j}$. Therefore

$$\operatorname{dist}(T^j(s_q), \mathbb{Z}) \geq \frac{1}{3 \cdot 2^{q-j}}.$$

The elementary inequality $|\sin \pi t| \geq 2 \operatorname{dist}(t, \mathbb{Z})$ now gives

$$\prod_{j=0}^{q-1} |\sin(\pi T^j(s_q))| \geq \prod_{j=0}^{q-1} \frac{2}{3 \cdot 2^{q-j}} = \left(\frac{2}{3}\right)^q 2^{-q(q+1)/2}.$$

After time q , the orbit alternates between $1/3$ and $2/3$, so every remaining factor equals $\sqrt{3}/2 = e^\beta$. Taking logarithms and absorbing the linear terms into Cq^2 proves (41). $\square$

The quadratic loss in q is more than sufficient: later we choose $q \asymp \log \log n$, while the main boundary-layer loss is of order $(\log n)^2$.

6.3 The finite-dimensional objective

Combining (38) with the tail rate β , define

$$\mathcal{F}_n(r, s) = -\frac{\lambda}{2}r^2 + r \left(\log(\pi s) - \frac{\lambda}{2} - \beta \right) - n \log(1 + s2^{-r}), \quad (42)$$

for integers $r \geq 0$ and $s \in I$.

Lemma 6.2 (Reduction to $\mathcal{F}_n$). *As $n \rightarrow \infty$,*

$$\log A_n = n\beta + \sup_{\substack{r \in \mathbb{Z}_{\geq 0} \\ s \in I}} \mathcal{F}_n(r, s) + \mathcal{O}((\log \log n)^2). \quad (43)$$

Proof. We first record a benchmark. Let $r_0 = \lfloor (\log n)/\lambda \rfloor$, put $\theta_0 = 2^{-r_0}/3$, and use $Q_{n-r_0}(1/3) = e^{(n-r_0)\beta}$ in (36). The uniform bounds on H and H_{r_0} then give

$$\log A_n \geq n\beta - C(\log n)^2 \quad (44)$$

for an absolute constant C .

For the upper bound, first note that $0 \leq H(1 + \theta) \leq 1$ on $[0, 1]$. If $\theta \geq 1/2$, then

$$H(1 + \theta)(1 + \theta)^{-n} Q_n(\theta) \leq C_G \exp\left(n\beta - n \log \frac{3}{2}\right),$$

which is exponentially smaller than the benchmark (44). If $0 < \theta < 1/2$, use (35). When $r < n$, (37) and (39) give the right-hand side of (43), with an $\mathcal{O}(1)$ error. When $r \geq n$, all n sine factors are in their small-argument regime, and

$$Q_n(\theta) \leq (\pi\theta)^n 2^{n(n-1)/2} \leq \exp\left(-\frac{\lambda}{2}n^2 + \mathcal{O}(n)\right),$$

again negligible. This proves the upper bound.

For the lower bound, choose (r, s_0) within 1 of the supremum of $\mathcal{F}_n$. We first record a quantitative localization. With $r_0 = \lfloor (\log n)/\lambda \rfloor$ and $s = 1/3$, one has $s2^{-r_0} = \mathcal{O}(n^{-1})$, and therefore

$$\mathcal{F}_n(r_0, 1/3) \geq -C_1(\log n)^2$$

for an absolute constant C_1 . On the other hand, because $\log(\pi s) - \lambda/2 - \beta$ is bounded on I ,

$$\mathcal{F}_n(r, s) \leq -\frac{\lambda}{2}r^2 + C_2r.$$

Comparison with the preceding lower bound forces $r = \mathcal{O}(\log n)$ at every near-maximizer. Since $0 \leq s2^{-r} \leq 1/2$ and $\log(1 + u) \geq u/2$ on this interval, we also have

$$\mathcal{F}_n(r, s) \leq C_3r - \frac{1}{2}ns2^{-r}.$$

A second comparison with $-C_1(\log n)^2$ now gives

$$r = \mathcal{O}(\log n), \quad ns_0 2^{-r} = \mathcal{O}((\log n)^2). \quad (45)$$

Take

$$q = \lceil 4 \log_2(\log(n + e^e)) \rceil$$

after increasing it by a fixed constant if necessary. For all sufficiently large n , the localization (45) implies $n - r \geq q$. Apply lemma 6.1 to obtain s_q with $|s_q - s_0| \leq 2^{-q}$. The derivative of $\mathcal{F}_n(r, s)$ with respect to s , throughout I , is bounded by $\mathcal{O}((\log n)^2)$ at the localized values in (45). Hence

$$\mathcal{F}_n(r, s_q) = \mathcal{F}_n(r, s_0) + \mathcal{O}(1).$$

The factors $H(1 + 2^{-r}s_q)$ and $H_r(s_q)$ are bounded below by positive constants. Finally, (41) supplies the tail with a loss $Cq^2 = \mathcal{O}((\log \log n)^2)$. This gives the lower bound in (43). $\square$

6.4 Lambert W optimization

The remaining optimization has a useful exact structure. Put

$$C_0 = \log \pi - \frac{\lambda}{2} - \beta = \log \left(\pi \sqrt{\frac{2}{3}} \right), \quad c = \lambda e^{-C_0} = \frac{\lambda}{\pi} \sqrt{\frac{3}{2}}. \quad (46)$$

Numerically,

$$c = 0.27022231973336536 \dots \quad (47)$$

Lemma 6.3 (Lambert optimization). *Let $w_n = W(cn)$, where W is the principal Lambert function. Then*

$$\sup_{\substack{r \in \mathbb{Z}_{\geq 0} \\ s \in I}} \mathcal{F}_n(r, s) = -\frac{w_n^2 + 2w_n}{2\lambda} + \mathcal{O}(1). \quad (48)$$

Proof. First replace $-n \log(1 + s2^{-r})$ by $-ns2^{-r}$, and denote the resulting function by $F_n(r, s)$. At a maximizer of $\mathcal{F}_n$, the localization argument in (45) gives $ns2^{-r} = \mathcal{O}((\log n)^2)$. Therefore

$$0 \leq n(s2^{-r} - \log(1 + s2^{-r})) \leq \frac{n}{2}s^22^{-2r} = \mathcal{O}\left(\frac{(\log n)^4}{n}\right) = o(1).$$

The same comparison applies to a maximizer of F_n , so the two suprema differ by $o(1)$.

Set

$$A = C_0 + \log s.$$

As s ranges over $I = [1/4, 1/2]$, the variable A ranges over an interval $[A_-, A_+]$ of length exactly $A_+ - A_- = \log 2 = \lambda$. Now put

$$z = \lambda r - A. \quad (49)$$

Since $s = e^{A-C_0}$, a direct calculation gives

$$F_n(r, s) = -\frac{z^2}{2\lambda} - ne^{-C_0}e^{-z} + \frac{A^2}{2\lambda}. \quad (50)$$

For successive integers r , the intervals of possible z in (49) have length λ and meet end to end. Thus they tile the real line, apart from an irrelevant lower truncation caused by $r \geq 0$. The last term in (50) is uniformly bounded. Consequently, the supremum differs by $\mathcal{O}(1)$ from the maximum over real z of

$$\phi_n(z) = -\frac{z^2}{2\lambda} - ne^{-C_0}e^{-z}.$$

The critical-point equation is

$$-\frac{z}{\lambda} + ne^{-C_0}e^{-z} = 0, \quad \text{or equivalently} \quad ze^z = \lambda ne^{-C_0} = cn.$$

Hence the unique maximizer is $z = w_n = W(cn)$. At that point, $ne^{-C_0}e^{-w_n} = w_n/\lambda$, so

$$\phi_n(w_n) = -\frac{w_n^2}{2\lambda} - \frac{w_n}{\lambda} = -\frac{w_n^2 + 2w_n}{2\lambda}.$$

This proves (48). □

The tiling argument is the reason the integer transition time r does not create an $\mathcal{O}(\log n)$ rounding error. The continuous variable s , whose allowed interval has multiplicative width exactly two, compensates for the lattice step in r . The remaining loss in lemma 6.2 comes from forcing the tail orbit to be genuinely near-maximizing.

6.5 Main envelope theorem

Theorem 6.4 (Sharp dyadic-annulus envelope). *Let*

$$M_k = \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|, \quad n = k + 1, \quad w_k = W(cn),$$

where c is given by (46). Then, as $k \rightarrow \infty$,

$$\log M_k = -\frac{\lambda}{2}k(k+1) - n \log \pi + n\beta - \frac{w_k^2 + 2w_k}{2\lambda} + \mathcal{O}((\log \log k)^2). \quad (51)$$

Equivalently, with a from (30),

$$\log M_k = -\frac{\lambda}{2}k^2 - ak - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2\lambda} + \mathcal{O}((\log \log k)^2). \quad (52)$$

Proof. Combine (34), lemma 6.2, and lemma 6.3. The constant difference between (51) and (52) is absorbed by the error term. $\square$

The Lambert form is more accurate and more informative than a truncated series. Expanding

$$W(cn) = \log n - \log \log n + \log c + \mathcal{O}\left(\frac{\log \log n}{\log n}\right) \quad (53)$$

gives the following explicit hierarchy.

Corollary 6.5 (Iterated-log expansion). *As $k \rightarrow \infty$,*

$$\begin{aligned} \log M_k = & -\frac{\lambda}{2}k^2 - ak - \frac{(\log k)^2}{2\lambda} + \frac{\log k \log \log k}{\lambda} \\ & - \frac{1 + \log c}{\lambda} \log k + \mathcal{O}((\log \log k)^2). \end{aligned} \quad (54)$$

In terms of the physical frequency scale, let $R = 2^k$, $L = \log R$, and

$$d = \frac{1}{\pi} \sqrt{\frac{3}{2}}. \quad (55)$$

Then $c(k+1) = d(L + \lambda)$, and

$$\begin{aligned} \log \max_{R \leq x \leq 2R} |\Phi(x)| = & -\frac{L^2}{2\lambda} - \frac{a}{\lambda}L \\ & - \frac{W(d(L + \lambda))^2 + 2W(d(L + \lambda))}{2\lambda} + \mathcal{O}((\log \log L)^2). \end{aligned} \quad (56)$$

Since $W(dL) \sim \log L = \log \log R$, the first terms of the additional boundary-layer correction are

$$-\frac{(\log \log R)^2}{2\lambda} + \frac{\log \log R \log \log \log R}{\lambda} + \mathcal{O}(\log \log R). \quad (57)$$

This is the decisive correction missed by a normalization containing only a fixed power of $\log R$.

6.6 Geometry of a near-maximizing point

The proof also explains where the large peak lies. At the smooth Lambert optimum, the variable z is w_n , and

$$n\theta = ns2^{-r} = ne^{-C_0}e^{-z} = \frac{w_n}{\lambda}.$$

The preimage approximation in [lemma 6.1](#) changes this only by lower-order factors. Thus one can construct near-maximizing points with

$$\theta \asymp \frac{W(cn)}{n}, \quad r = \frac{W(cn)}{\lambda} + \mathcal{O}(1) = \log_2 n - \log_2 \log n + \mathcal{O}(1). \quad (58)$$

The frequency therefore has the form

$$x = 2^k \left(1 + \Theta \left(\frac{\log k}{k} \right) \right). \quad (59)$$

The orbit spends r steps in the small-angle region, then enters a compact part of the circle and shadows an eventually periodic preimage of the maximizing cycle. The cost of the first phase is quadratic in r ; the endpoint penalty is of order $n2^{-r}$. Balancing these two costs is exactly the equation solved by Lambert W .

7 Two-adic troughs in the sequence of local peaks

The dyadic-annulus maximum selects the best peak among exponentially many unit intervals. The individual peak sequence E_n is much less regular. The most dramatic exceptions occur immediately after integers with large two-adic valuation.

7.1 Exact factorization near an odd multiple of a power of two

Fix an odd positive integer m , put

$$N = m2^k, \quad x = N + t, \quad 0 \leq t \leq 1,$$

and define the finite product

$$\Phi_k(t) = \prod_{j=0}^k \operatorname{sinc} \left(\frac{\pi t}{2^j} \right). \quad (60)$$

For $0 \leq j \leq k$, the integer $N/2^j$ contributes only a sign to the sine. Taking absolute values gives

$$\left| \operatorname{sinc} \left(\frac{\pi(N+t)}{2^j} \right) \right| = \frac{t}{N+t} \operatorname{sinc} \left(\frac{\pi t}{2^j} \right).$$

The remaining factors are the tail $H(m + t/2^k)$. Hence the exact identity

$$|\Phi(N+t)| = \left(\frac{t}{N+t} \right)^{k+1} \Phi_k(t) \left| H \left(m + \frac{t}{2^k} \right) \right|. \quad (61)$$

Because m is odd, $H(m) \neq 0$.

As $k \rightarrow \infty$, uniformly for $0 \leq t \leq 1$,

$$\Phi_k(t) = \Phi(t)(1 + \mathcal{O}(4^{-k})), \quad H \left(m + \frac{t}{2^k} \right) = H(m)(1 + \mathcal{O}(2^{-k})), \quad (62)$$

The first estimate follows by writing $\Phi = \Phi_k R_k$, where the omitted tail R_k is positive on $[0, 1]$ and satisfies $R_k = 1 + \mathcal{O}(4^{-k})$; thus the identity remains valid at $t = 1$, where both sides vanish. The second estimate follows from analyticity and $H(m) \neq 0$. Finally,

$$\left(1 + \frac{t}{N}\right)^{-(k+1)} = 1 + o(1). \quad (63)$$

Thus the peak problem reduces to maximizing $t^p \Phi(t)$, where $p = k + 1$ tends to infinity.

7.2 A boundary maximum lemma

Lemma 7.1 (Linear zero against a large power). *Let f be continuous on $[0, 1]$, positive on $[0, 1)$, and suppose*

$$f(t) = h(1 - t) + \mathcal{O}((1 - t)^2) \quad (t \uparrow 1), \quad (64)$$

with $h > 0$. If

$$J_p = \max_{0 \leq t \leq 1} t^p f(t),$$

then

$$J_p \sim \frac{h}{ep}. \quad (65)$$

Moreover, if t_p is any maximizing point, then

$$p(1 - t_p) \longrightarrow 1. \quad (66)$$

Proof. Evaluation at $t = 1 - 1/p$, together with (64), gives

$$J_p \geq \left(1 - \frac{1}{p}\right)^p \left(\frac{h}{p} + \mathcal{O}(p^{-2})\right) = \frac{h + o(1)}{ep}. \quad (67)$$

For each fixed $\varepsilon > 0$, the part $0 \leq t \leq 1 - \varepsilon$ contributes at most $(1 - \varepsilon)^p \|f\|_\infty = o(p^{-1})$. Hence every maximizing point t_p tends to 1.

Put $s = p(1 - t)$. Uniformly for s in a fixed compact subset of $[0, \infty)$,

$$p t^p f(t) \longrightarrow h s e^{-s}. \quad (68)$$

It remains only to prevent the maximizing scaled variable $s_p = p(1 - t_p)$ from escaping compact sets. For t sufficiently close to 1, (64) gives $f(t) \leq C(1 - t)$, while $t^p \leq \exp(-p(1 - t))$. Therefore

$$p t^p f(t) \leq C s e^{-s}. \quad (69)$$

The right-hand side tends to zero both as $s \downarrow 0$ and as $s \rightarrow \infty$, whereas (67) supplies a positive limiting benchmark. Thus s_p remains in a fixed compact subset of $(0, \infty)$. Every subsequential limit of s_p must, by (68), maximize $h s e^{-s}$. This maximum is unique and occurs at $s = 1$. Hence $s_p \rightarrow 1$, and the maximum value is asymptotic to $h/(ep)$. $\square$

For $f = \Phi$, formula (6) gives

$$\Phi(t) = h(1 - t) + \mathcal{O}((1 - t)^2), \quad h = H(1), \quad (70)$$

because $(\text{sinc}(\pi t))'|_{t=1} = -1$.

Theorem 7.2 (Peaks following $m2^k$). *Fix an odd positive integer m . Let $c_{m2^k} = m2^k + t_k$ be the unique critical point in $(m2^k, m2^k + 1)$. Then*

$$(k+1)(1-t_k) \longrightarrow 1, \quad (71)$$

and

$$E_{m2^k} \sim \frac{H(1)|H(m)|}{e(k+1)} (m2^k)^{-(k+1)}. \quad (72)$$

In particular,

$$E_{2^k} \sim \frac{H(1)^2}{e(k+1)} 2^{-k(k+1)}. \quad (73)$$

Proof. Let $p = k+1$. From (61)–(63), uniformly in $t \in [0, 1]$,

$$|\Phi(m2^k + t)| = (m2^k)^{-p} |H(m)| t^p \Phi(t) (1 + o(1)).$$

Apply lemma 7.1 with $f = \Phi$ and (70). The multiplicative error above tends to one uniformly, so the benchmark and compactness argument in that lemma apply unchanged to the perturbed maximization problem. The uniqueness of the maximizer follows from theorem 2.1, and the value and location asymptotics follow. $\square$

Corollary 7.3 (Log-normal spread inside one annulus). *The peak immediately after 2^k is exponentially smaller on the log-normal scale than the largest peak in $[2^k, 2^{k+1}]$:*

$$\log \frac{E_{2^k}}{M_k} = -\frac{\lambda}{2} k^2 + \mathcal{O}(k). \quad (74)$$

In particular, $E_{2^k}/M_k \rightarrow 0$ faster than every exponential in k .

Proof. From (73), $\log E_{2^k} = -\lambda k(k+1) - \log(k+1) + \mathcal{O}(1)$. Subtract (52). $\square$

This result rules out not only a pointwise constant-amplitude normalization, but also any model in which all successive interzero peaks are assumed to follow one smooth asymptotic curve. The two-adic valuation remains visible at arbitrarily large frequencies.

8 What can and cannot be normalized

8.1 Pointwise normalization

No positive function $g(x)$ can satisfy

$$\frac{|\Phi(x)|}{g(x)} \longrightarrow C > 0 \quad (x \rightarrow \infty)$$

pointwise, because the quotient is zero at every positive integer. Replacing all points by local maxima does not restore a single regular law: corollary 7.3 shows a log-normal spread among peaks whose frequencies differ by a factor less than two.

What does exist is a hierarchy of precise normalizations:

Object	Rigorous decay law
Universal limsup	$\log \Phi(x) \sim_{\limsup} -(\log x)^2 / (2 \log 2)$.
Typical dyadic ray	$\log \Phi(x) = -(\log x)^2 / (2 \log 2) - (3/2 + \log \pi / \log 2) \log x + o(\log x)$.
Maximizing fixed ray	The same quadratic term, with linear coefficient $-a / \log 2$, attained by the orbit $\{1/3, 2/3\}$.
Dyadic-annulus maximum	The Lambert W formula (51), including a second log-normal correction in $\log \log x$.
Peak after $m2^k$	The much smaller two-adic law (72).

8.2 Why the draft's proposed second term is not valid

The draft [7] writes, in effect,

$$\log |\Phi(x)| \stackrel{?}{=} -\frac{(\log x)^2}{2\lambda} - \frac{\log(2\pi)}{\lambda} \log x - \frac{1}{2} \log \log x + \mathcal{O}(1). \quad (75)$$

There are several independent obstructions.

1. At the integer zeros, the left side is $-\infty$.
2. On almost every dyadic ray, the true linear coefficient is $-3/2 - \log \pi/\lambda$, not $-\log(2\pi)/\lambda$; compare (23).
3. On maximizing rays the linear coefficient is again different; compare (30).
4. For the largest value in a dyadic annulus, the correction after the k^2 and k terms is

$$-\frac{(\log k)^2}{2\lambda} + \frac{\log k \log \log k}{\lambda} + \dots,$$

or, in the original frequency variable, the second log-normal term (57). It cannot be represented by $-p \log \log x$ with constant p .

5. The peak sequence has the exceptional subsequence (72), whose leading quadratic coefficient in k is twice as negative as that of M_k .

The Fourier-series identity for $\log |\sin|$ is useful only after the orbit structure and its singularities are controlled. Replacing the dyadic orbit by its mean without specifying whether the problem is typical, extremal, or averaged loses precisely the information that determines the lower-order terms.

9 The original Cesaro-average question is a different problem

The Mathematics Stack Exchange question that motivated the draft asks for a positive function g such that a Cesaro mean of $|\Phi|/g$ is bounded or convergent [6]. The present L^∞ analysis does not by itself determine that normalization.

Indeed, integrating (32) gives

$$\int_{2^k}^{2^{k+1}} |\Phi(x)| dx = 2^k \pi^{-n} 2^{-k(k+1)/2} I_n, \quad (76)$$

$$I_n = \int_0^1 H(1+\theta)(1+\theta)^{-n} Q_n(\theta) d\theta.$$

The quantity I_n is a weighted L^1 norm of a Thue–Morse product. By (18), the unweighted version is proportional to the L^1 norm of the length- 2^n Thue–Morse polynomial, a separate and subtle object discussed in the harmonic-analysis literature [5]. Its asymptotics are governed by pressure or transfer-operator information, not solely by the Gelfond L^∞ exponent.

Thus the sharp envelope derived here supplies an upper constraint and identifies the rare configurations producing the largest spikes, but a Cesaro normalizer requires additional information about the measure and width of near-maximizing orbit sets. Conflating these two questions is one source of the incorrect pointwise claim in (75).

10 Numerical verification

The proofs above are analytic. Numerical computation is nevertheless useful for checking normalizations and for showing how early the Lambert formula becomes accurate.

For each integer j , [theorem 2.1](#) permits a one-dimensional maximization of

$$\sum_{h=0}^H \log \left| \operatorname{sinc} \left(\frac{\pi x}{2^h} \right) \right|$$

on $(j, j+1)$. A tail bound from $\log \operatorname{sinc} u = -u^2/6 + \mathcal{O}(u^4)$ controls the truncation. Taking the largest of these unit-interval peaks produces M_k . The table compares the computed value with the centered Lambert approximation

$$\mathcal{A}_k = -\frac{\lambda}{2}k(k+1) - (k+1)\log \pi + (k+1)\beta - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2\lambda}. \quad (77)$$

k	maximizing $x/2^k$	$\log M_k$	$\mathcal{A}_k$	residual
3	1.177288	-11.251778	-10.428644	-0.823133
4	1.162272	-15.485667	-14.695388	-0.790279
5	1.168960	-20.390823	-19.639196	-0.751626
6	1.165501	-25.991031	-25.262965	-0.728066
7	1.167235	-32.287929	-31.568814	-0.719116
8	1.166376	-39.274011	-38.558353	-0.715658
9	1.166810	-46.956338	-46.232839	-0.723499
10	1.166594	-55.329708	-54.593276	-0.736432
11	1.166703	-64.397553	-63.640479	-0.757074
12	1.166649	-74.157742	-73.375125	-0.782618

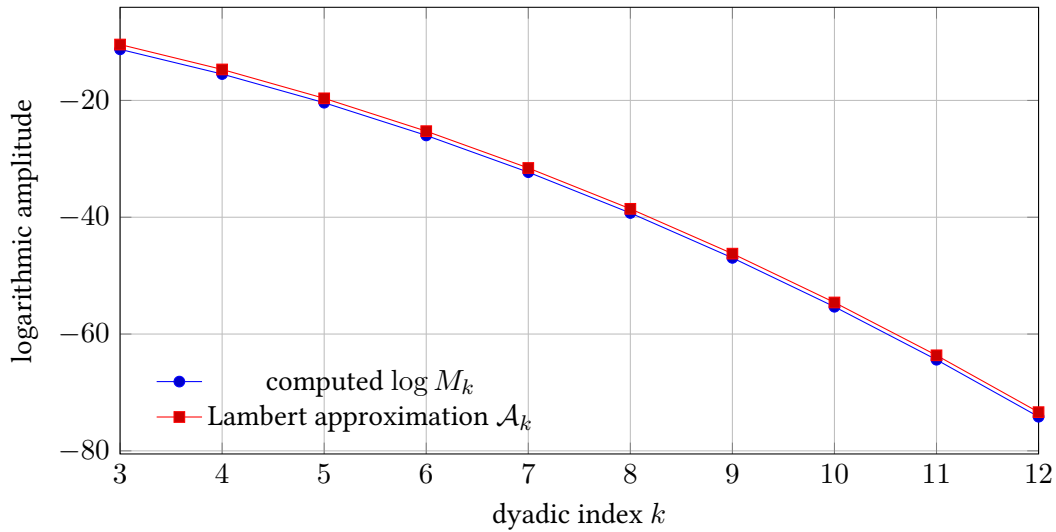


Figure 1: The centered Lambert formula tracks the annular maximum with a small bounded-size residual even at modest values of k . The theorem only claims an $\mathcal{O}((\log \log k)^2)$ remainder.

For the displayed range, the maximizing ratio is close to $7/6$. This corresponds to the simple preperiodic choice $\theta = 1/6$, which reaches the maximizing cycle after one doubling. At much larger k , the optimal preperiod grows according to (58); the convergence is necessarily slow because it is only logarithmic in k .

11 Conclusions and a formalization roadmap

The real-axis Fourier decay of the up-function is not described by one smooth amplitude curve. It has a clean but layered structure.

1. The coefficient of $(\log x)^2$ is universally sharp in the limsup sense and equals $-1/(2 \log 2)$.
2. Fixed dyadic rays are controlled by Birkhoff averages of $\log |\sin \pi x|$. Almost every ray has average $-\log 2$, whereas the largest possible average is $\log(\sqrt{3}/2)$.
3. The sharp Gelfond rate follows from the elementary polynomial identity (28); no computer-assisted ergodic optimization is needed in this special case.
4. The largest peak in a dyadic annulus is a boundary-layer phenomenon. Its exact asymptotic through an $\mathcal{O}((\log \log k)^2)$ remainder is the Lambert formula (51).
5. The first correction beyond the main log-normal envelope is itself log-normal in $\log x$, namely quadratic in $\log \log x$. A fixed factor $(\log x)^{-p}$ cannot capture it.
6. Individual local peaks retain strong two-adic irregularity. The peaks after $m2^k$ obey (72) and are vastly smaller than the annular maximum.

The proof was organized to be formalization-friendly. A Lean development can be divided into the following relatively independent blocks:

1. locally uniform convergence, the scaling law, and the canonical zero product;
2. monotonicity of the logarithmic-derivative sum and uniqueness of the interzero peak;
3. the exact dyadic-ray and annular decompositions;
4. the algebraic identity (28) and its telescoping consequence;
5. the rational preimage construction in lemma 6.1;
6. elementary localization estimates and the Lambert optimization in lemma 6.3;
7. uniform convergence in (62) and the boundary maximum lemma.

The only general ergodic input needed for the almost-everywhere statement is Birkhoff's theorem. The main extremal envelope and the two-adic peak theorem are otherwise elementary consequences of the product itself.

A Expanded forms and constants

For convenience, the constants occurring in the main theorem are

$$\begin{aligned}
 \lambda &= \log 2 = 0.6931471805599453 \dots, \\
 \beta &= \log(\sqrt{3}/2) = -0.1438410362258904 \dots, \\
 a &= \frac{\lambda}{2} + \log \pi - \beta = 1.6351445123552633 \dots, \\
 c &= \frac{\lambda}{\pi} \sqrt{\frac{3}{2}} = 0.27022231973336536 \dots, \\
 d &= \frac{1}{\pi} \sqrt{\frac{3}{2}} = 0.3898484006168381 \dots, \\
 H(1) &= 0.55377127588881009534 \dots
 \end{aligned}$$

Let $L_1 = \log k$ and $L_2 = \log \log k$. From the standard expansion of Lambert W [2],

$$W(c k) = L_1 - L_2 + \log c + \frac{L_2 - \log c}{L_1} + \mathcal{O}\left(\frac{L_2^2}{L_1^2}\right).$$

Inserting only the terms justified above the remainder in theorem 6.4 gives (54). Further formal terms can be written, but the present proof's orbit-shadowing error is $\mathcal{O}(L_2^2)$, so coefficients at that order are not claimed.

B A compact proof of rapid decrease

For completeness, fix $N \geq 1$. From the first N factors of the product,

$$|x|^N |\Phi(x)| \leq |x|^N \prod_{h=0}^{N-1} \frac{2^h}{\pi|x|} = \frac{2^{N(N-1)/2}}{\pi^N}.$$

Thus Φ is rapidly decreasing. Differentiating the product and using the same estimate, together with the geometric factors introduced by each derivative, proves that every derivative of Φ is rapidly decreasing; see [1] for the complete Schwartz-space argument.

References

- [1] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation and expanded version, arXiv:1702.05442, 2017. <https://arxiv.org/abs/1702.05442>.
- [2] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359. <https://doi.org/10.1007/BF02124750>.
- [3] A. Fan, J. Schmeling, and W. Shen, L^∞ -estimation of generalized Thue–Morse trigonometric polynomials and ergodic maximization, *Discrete and Continuous Dynamical Systems* **41** (2021), no. 1, 297–327. <https://doi.org/10.3934/dcds.2020363>.
- [4] A. O. Gelfond, Sur les nombres qui ont des proprietes additives et multiplicatives donnees, *Acta Arithmetica* **13** (1967/68), 259–265. <https://doi.org/10.4064/aa-13-3-259-265>.
- [5] M. Queffelec, Questions around the Thue–Morse sequence, *Uniform Distribution Theory* **13** (2018), no. 1, 1–25. <https://doi.org/10.1515/udt-2018-0001>.
- [6] V. Reshetnikov, Asymptotic decay rate of an infinite product of sinc functions, Mathematics Stack Exchange, question 2745497, 20 April 2018. <https://math.stackexchange.com/q/2745497>.
- [7] V. Reshetnikov, *Decay rate conjecture*, non-formalized research-frontier draft in the ProveIt repository, consulted 26 August 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/Decay%20rate%20conjecture/Decay%20rate%20conjecture.tex>.
- [8] Wikipedia contributors, Fabius function, consulted 26 August 2026. https://en.wikipedia.org/wiki/Fabius_function.
- [9] P. Walters, *An Introduction to Ergodic Theory*, Graduate Texts in Mathematics 79, Springer, 1982.